Tabletop Steam Sterilizer

Installation & Users Manual

HäTmed (Wuxi) Co., Ltd.

Version: V01

Foreword

Thank you for using the HäTmed steam sterilizer, this manual covers installation, basic operations, maintenance and safety information of the tabletop steam sterilizer in details for your convenience, please read this manual carefully before use.

Explanation

- This manual applies to Qx0B/N, Qx1B/N models in Quantus series steam sterilizers.
- Models with N suffix (e.g.: Q70N) are sterilizers without pre-vacuum, for these models articles 4.1, 4.6, 5.3, and 7.9.2 are not applicable.

Terms

- STERILIZER: Quantus Tabletop steam sterilizers
- STERILIZATION LOAD: The items to be sterilized (packed or non-packed)
- STERILIZATION CHAMBER: the space that the sterilization loads are placed and sterilized
- HOT START: to start the sterilizer in less than one hour after the previous sterilization cycle when the chamber may still be warm

Installation & Users Manual

Signals

Signal	Information	Explanation
	Potential safety hazard	High voltage or large current near the signal. Avoid it carefully; otherwise your personal safety may be endangered.
Warning	Be sure to follow	Dangerous condition near the signal. Avoid it with care; otherwise the sterilization load or the sterilizer may be damaged.
Note	Be sure to follow	The message after the signal must be followed; otherwise cycle will fail to sterilize and the load or the sterilizer may be damaged.

Safety Information

Warning

- Please read the manual carefully before use.
- Carefully read the information after the signal **"Note"**.
- Do not sterilize any liquid.
- Do not replace the main power cord arbitrarily.
- Switch the power off before removing the plug.
- Do not start any operation if the power cord is damaged.
- Mind the steam while opening the door after operation.
- If sterilization stops before drying stage, process is invalid.
- In order to avoid scald, do not fetch the tray with bare hands.
- If error messages repeat in the sterilization, try rebooting machine.
- Repair and maintenance can only be carried out by qualified professionals.

Special Information

Note

- -1 Bar means the pressure value at absolute vacuum state.

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1 Product Description

1.1 Technical Parameters

Volume	18L,23L
Power	2500VA; AC230V±10%/16A; 50Hz±1Hz
Pressure/ Temperature	2.1 Bar/134°C; 1.1 Bar /121°C
Size（W x D x H）	See Table Below
Package size（W x D x H ）	See Table Below

Model	Size（W x D x H）	Package size（W x D x H ）
Q50B	462mm*452mm*469mm	688mm*576mm*688mm
Q51B	462mm*452mm*469mm	688mm*576mm*688mm
Q50N	462mm*452mm*469mm	688mm*576mm*688mm
Q51N	462mm*452mm*469mm	688mm*576mm*688mm
Q52N	567mm*454mm*436mm	688mm*576mm*688mm
Q70B	682mm*452mm*469mm	806mm*576mm*688mm
Q71B	682mm*452mm*469mm	806mm*576mm*688mm
Q70N	682mm*452mm*469mm	806mm*576mm*688mm
Q71N	682mm*452mm*469mm	806mm*576mm*688mm
Q72N	686mm*455mm*434mm	806mm*576mm*688mm

1.2 Applications

Classified by profession: For medical, dental clinical, beauty salon and laboratory use.

Classified by the sterilization load: Class B and Class N. Classified as B class, applied to harsh sterilization requirements such as hollow and porous loads, packed or non-packed and a large number of textile fabrics. Class N is applied to a sterilization load consisting of unpacked solid metal instruments.

Warning

- Liquids are not to be sterilized as damage will occur to sterilizer and load
- Do not use this equipment for any purpose other than its intended use described in this manual.
- Only sterilization loads permitted by their manufacturers for processing using suitable cycles may be processed.

2 Connectors and Components

2.1 Views of Connectors and Components

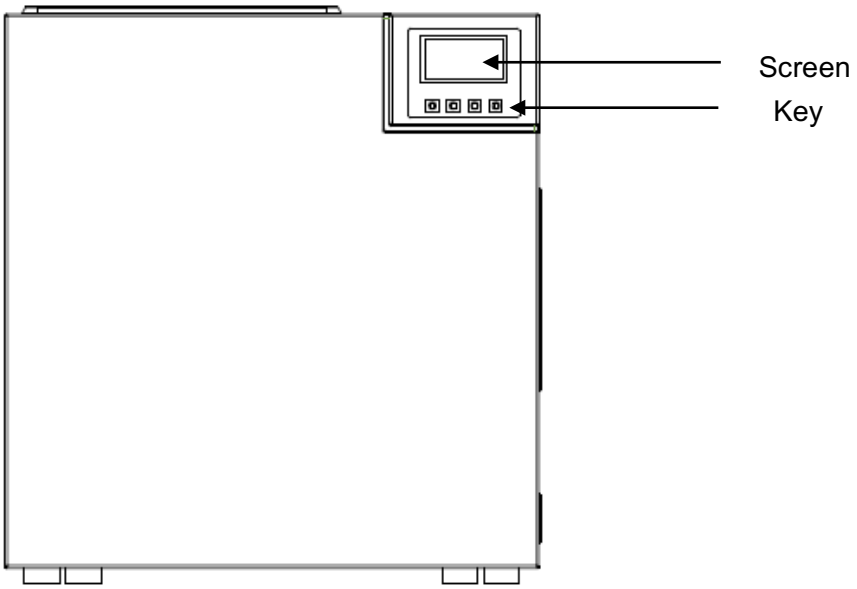


Figure 2-1 Front view

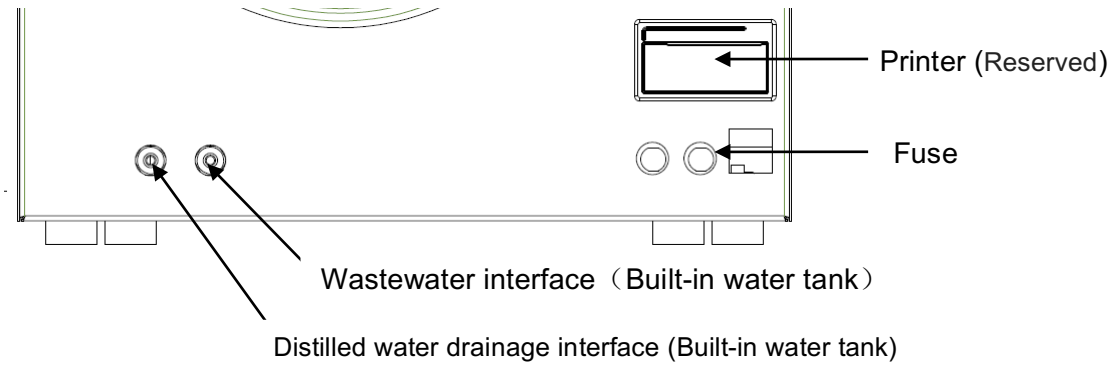


Figure 2-2 Internal view

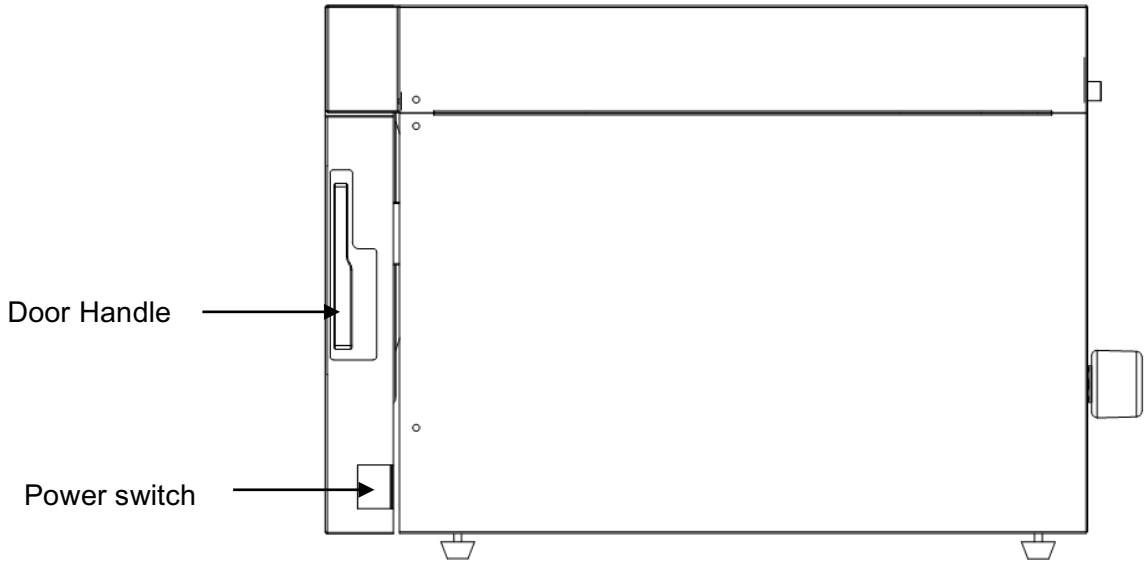


Figure 2-3 Side view

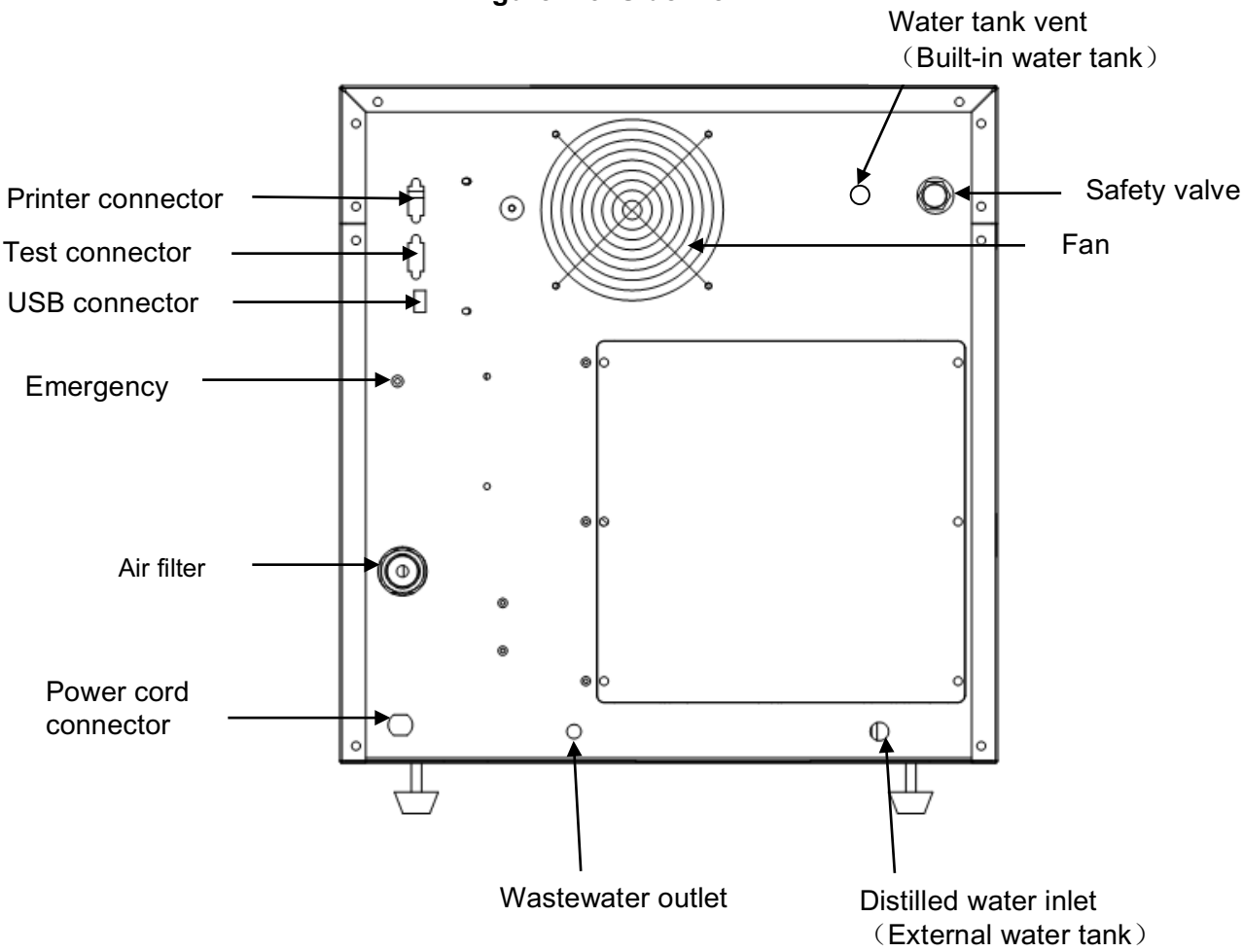


Figure 2-4 Back view

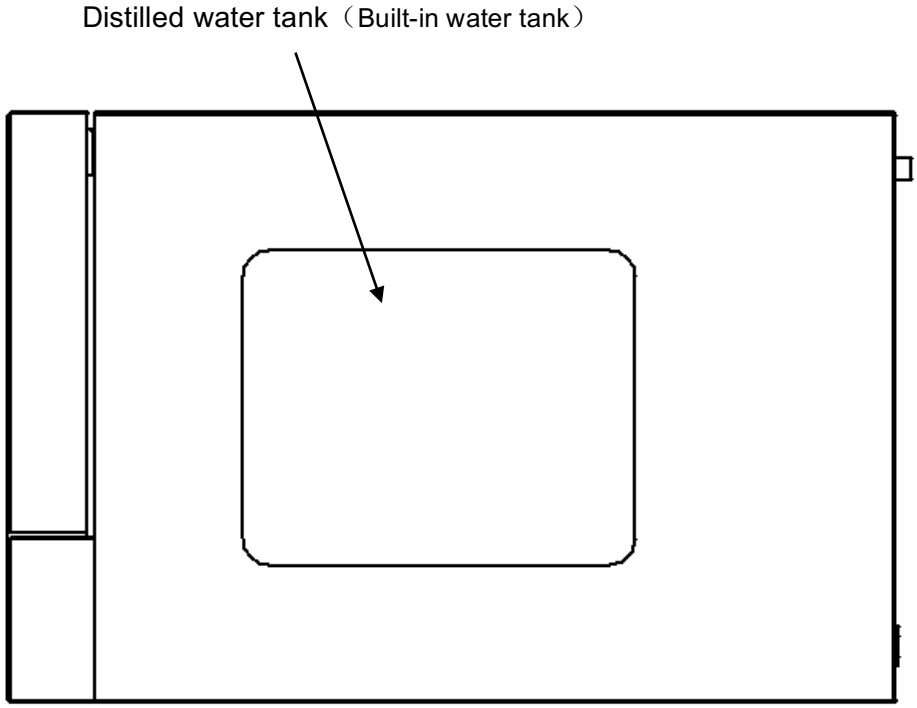


Figure 2-5 Top view

2.2 Connectors and Components

2.2.1 Screen

The screen is an LCD. The friendly interface and simple operation offer you the most convenient experience.

2.2.2 Door Handle

Door handle is specially designed for your convenience. In the main screen, Press the V key, the lock sign at bottom left of the screen changes to unlock sign indicating the door can be opened. Then pull the right side of the sterilizer door handle to about 90° to open the door.

2.2.3 USB Connector

USB connector is a data channel through which sterilization records can be exported to a computer or terminal.

2.2.4 Test Port

The test connector is the data interface for factory test or upgrading the code version.

2.2.5 Fuse

Two 230V 16A fast-blow fuses are the main power supply fuses. To change fuses, use a suitable screwdriver to remove the fuse cover. Tighten the cover after replacing the fuses.

2.2.6 Printer Port

A printer may be connected to the sterilizer via this port to enable printing of the sterilization records. If your sterilizer comes with an external printer, you can connect the data line and the power cord to the printer and print the sterilization records.

2.2.7 Safety Valve

Safety valve is an essential safety protection device in the sterilizer. When the pressure exceeds 1.3 times of the working pressure in the chamber, the safety valve opens automatically, and release the pressure in the chamber in time to avoid dangers.

2.2.8 Air Filter

The air filter is for filtering air into the chamber.

2.2.9 Wastewater Outlet (External water tank)

Wastewater outlet is for draining the wastewater produced in the sterilization process.

Insert drain-pipe into the wastewater outlet at the bottom of the sterilizer (as shown in Figure 2.4).

2.2.10 Distilled Water Inlet (External water tank)

The distilled water inlet is for providing distilled water required during sterilization.

2.2.11 Wastewater outlet (Built-in water tank)

Find the drain plug and drain pipe in the accessory bag. Connect them through the end without the black seal ring. Insert the plug of the wastewater outlet at the back of the device through the end with the black seal ring. Wastewater will automatically be drained. There are two wastewater outlets in the sterilizer, one in front, as shown in Figure 2.2, the other at back.

2.2.12 Distilled water outlet (Built-in water tank)

The distilled water outlet is in the front, as shown in Figure 2.2. Use the same connector and pipe for draining from wastewater outlet to drain water from distilled water tank.

2.2.13 Distilled water tank (Built-in water tank)

Models with built-in water tank come with distilled water tank, located at the top of the machine.

2.2.14 Wastewater tank vent (Built-in water tank)

For models with built-in water tank and wastewater tank, there is a vent for air exhaust at the back to release pressure in the wastewater tank.

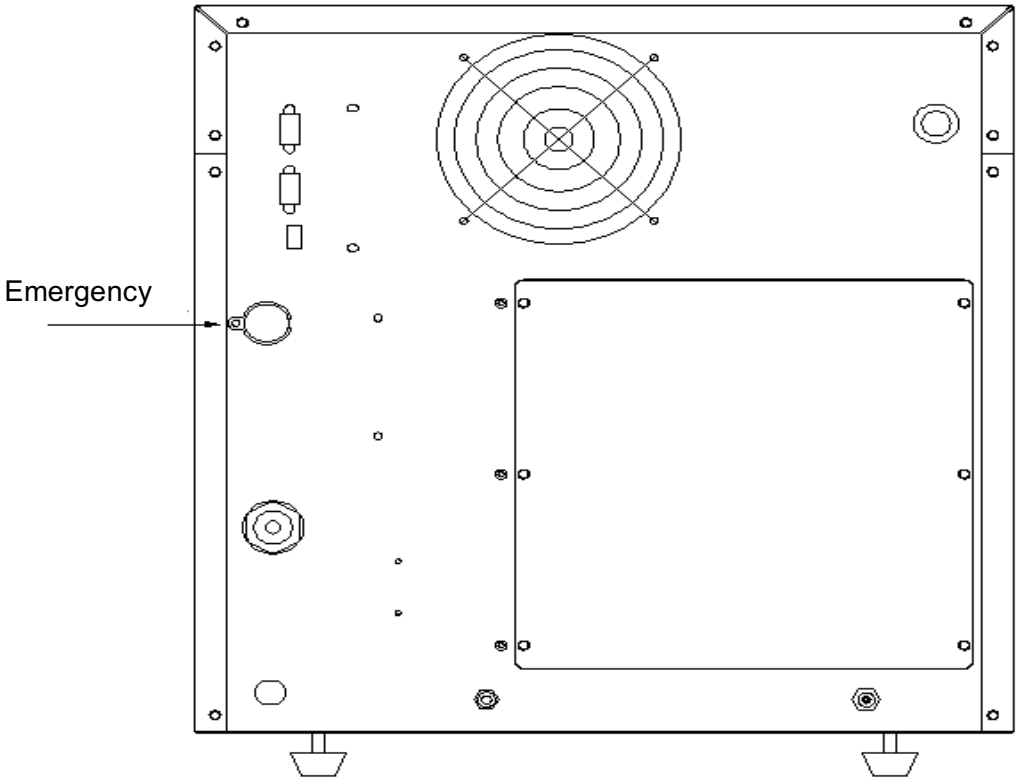
2.2.15 Emergency Door Handle

The emergency door handle at the back of the device must only be used in an emergency event. For example, during an electrical power failure or when something is wrong with the safety door lock system.

Do not use this handle under normal circumstances.

Care must be taken to ensure there is no pressure in the chamber before operating this handle.

Operation of emergence handle: uninstall the screw on the Emergency cap, pull gently on the button approximately 1 cm to unlock and then open the door with the door handle at the right side of the door.



2.2.16 Power Cord Connector

Warning

- The power cord connector is designed with an anti-pulled protection device. Do not pull hard when using it.

3 Installation and Transportation

3.1 Installation Environment

3.1.1 Installation Environmental Parameter

Sterilizer should be placed in a dust-free, dry environment. The installation environmental parameters are as following:

Working voltage	AC230V±10%/16A
Maximum power	2500W
Pressure	70 Kpa -106 Kpa
Temperature	5 ℃ – 40 ℃
Humidity	≤85%

3.1.2 Installation Site

Note

- Worktable: Must be able to support the weight of the sterilizer（40kg when the sterilizer is empty）. After placing the sterilizer, there should be space with at least 10 cm around and high above the sterilizer.
- Power plug: There should be a 16A power connector around.

3.2 Installation of Sterilizer

3.2.1 Unpacking Precautions

Be sure to check whether the carton is undamaged before unpacking, whether the anti-poured label on the right is failed. If there is any damage during the transportation, please contact the distributor immediately.

After opening the carton, take out the accessory box, and pull the carton up.

3.2.2 Accessories

Accessories with built-in tank:

No.	Accessory	Quantity	Description
1	Tray	3	Holders for sterilization loads to be placed in order
2	Steel wire frame	1	The frame to place the tray on
3	Tray handle	1	The tool to take out the tray.
4	Air filter	1	Air filter
5	Fuse	2	230V/16A.
6	Gradient	1	Ensure the front of the sterilizer is higher than the back.
7	Drain pipe	1	The pipe to drain the wastewater in the sterilizer
8	Drain plug	1	For connecting the drain pipe and the rear of the sterilizer

Accessories with external tank:

No.	Accessory	Quantity	Description
1	Tray	3	Holders for sterilization loads to be placed in order
2	Steel wire frame	1	The frame to place the tray on.
3	Tray handle	1	The tool to take the tray.
4	Air filter	1	Air filter
5	Fuse	2	230V/16A.
6	Gradient	1	Ensure the front of the sterilizer is higher than the back.
7	Drain pipe	1	The pipe to drain the wastewater in the sterilizer
8	Distilled water pipe	1	For supplying distilled water to the sterilizer

3.2.3 Foot Pads Height Adjustment

Take the gradient out of the accessory box and place it horizontally on the top of the sterilizer, adjust the foot pads at the front to make the front of the sterilizer higher than the back. This positioning is beneficial for the drying of the sterilization loads, and can save distilled water.

3.2.4 Power Connection

Connect to an AC230V/16A power socket. A dedicated 16A power supply is recommended in order to avoid power overload.

3.2.5 Printer Connection

If your sterilizer is installed with a printer, and you would like to print the sterilization records, take the printer pack out of the accessory box, take out the power cord and data line, connect the data line to the printer connector (**Figure 2-4**) at the back of the sterilizer, tighten up the screws, connect the other end to the printer data connector. The printer comes with a dedicated power supply. Load the printing paper after connecting the data line and power cord, switch on the power switch at the back of the printer (For the installation and the operation of the printer, please read the instructions of the printer).

Note

- It is recommended that all water pipes connected at the back of the sterilizer are downward and fixed firmly, ensure that the pipes are not folded or knotted.

3.2.6 The Use of Distilled Water

Warning

- Distilled water required by the sterilization should be high quality distilled or demineralized water. Bad quality distilled water leads to rust stains inside the sterilizer, so the distilled water used by the sterilizer should be homemade or bought from qualified supplier.
- For devices with water tank, when the water level inside the distilled water tank is too low or when the machine prompts lack of distilled water, it means the distilled water in water tank is not enough for one sterilization cycle and the sterilization will not work properly. For normal use, before starting sterilization please make sure there is sufficient distilled water in the water tank.
- Distilled water tank needs to be cleaned regularly, usually once a month.

3.3 Transportation

Remove all the pipes (cooling water pipe, distilled water pipe and wastewater pipe) at the back of the sterilizer before moving it. If the printer is connected, remove the data line.

If the frame and the trays are still in the chamber during transportation, put foams between the door and the frame before closing the door in order to prevent damage during transportation.

If there is long distance transportation, the sterilizer should be put back in the carton provided by HäTmed, and the carton should be sealed.

4 Features of HäTmed Sterilizer

4.1 Pulsating Vacuum

After starting the sterilization program, the vacuum system removes most of the air in the chamber during the first vacuuming. The following several vacuuming cycles fill steam into the chamber and then remove it, which ensures steam penetration inside the sterilization loads and the effectiveness of the sterilization. Meanwhile, pulsating vacuum can greatly improve the drying efficiency of the sterilization loads and reduce the sterilization time.

4.2 Sterilization Program

There are 6 common sterilization programs, 1 additional dry program and 3 test programs.

6 common sterilization programs:

Note

- GEN134 program: Mixed loads, complex devices such as long-shaped instruments with cavities.
- GEN121 program: A few textiles and heat-labile devices.
- Quick S program: Simple solid devices, transmission devices, instruments with simple straight cavity.
- Quick B program: A few packed devices or a large number of unpacked simple solid devices.
- Prion program: Heat-resistant devices that may be infected by Prion.
- Fabric program: A large number of fabric.

1 dry program:

- Dry program: With this program you can achieve optimal drying result for packed sterilization loads.

3 testing programs:

- Helix test program: Test program for steam penetration.
- B&D test program: Test program for the sterilization performance.
- Vacuum test program：Test program for any system leakage.

4.3 High Efficient Steam Generator

The special steam generator design generates steam quickly and ensures sufficient steam in the chamber during sterilization. The unique design allows accurate temperature control and prevents faults of the generator. The high efficiency results in reduced sterilization time and relatively small consumption of distilled water.

4.4 Economic Water Usage

There is an internal water cycling system which removes residual steam, condensation water and insoluble impurity after every sterilization, and provide a good environment for the next sterilization.

4.5 Intelligent Controller

The controller is the core of the sterilizer. It not only accomplishes real-time parameters acquisition, processing, fault detection and control of a sterilization process, but also intuitively depicts the graph of temperature and pressure that change over time.

The controller is also responsible for the assessment and monitoring of the program control process. During the sterilization period, it will compare the running parameter with the standard parameter, once the running parameter deviates from the standard, it will generate fault messages immediately, and ensure the safety and quality of the sterilization.

4.6 Vacuum Drying

The vacuum drying system has a very good drying effect, even for packed instruments.

4.7 Preheating Function

The preheating system can reduce cycle time and condensation time, thus improves the drying effect, and reduces sterilization time. If automatic preheating function is selected, it can preheat the chamber before the sterilization and maintain temperature in the chamber during sterilization, thus ensure quality of sterilization.

4.8 Sterilization Record

The system can store up to 60 sterilization records. Each record name is the cycle number. Users can download all or partial records with a USB disk and transfer them to a computer, or print the records you select with a printer. To print sterilization records of a particular day, just need to enter the date plus cycle number.

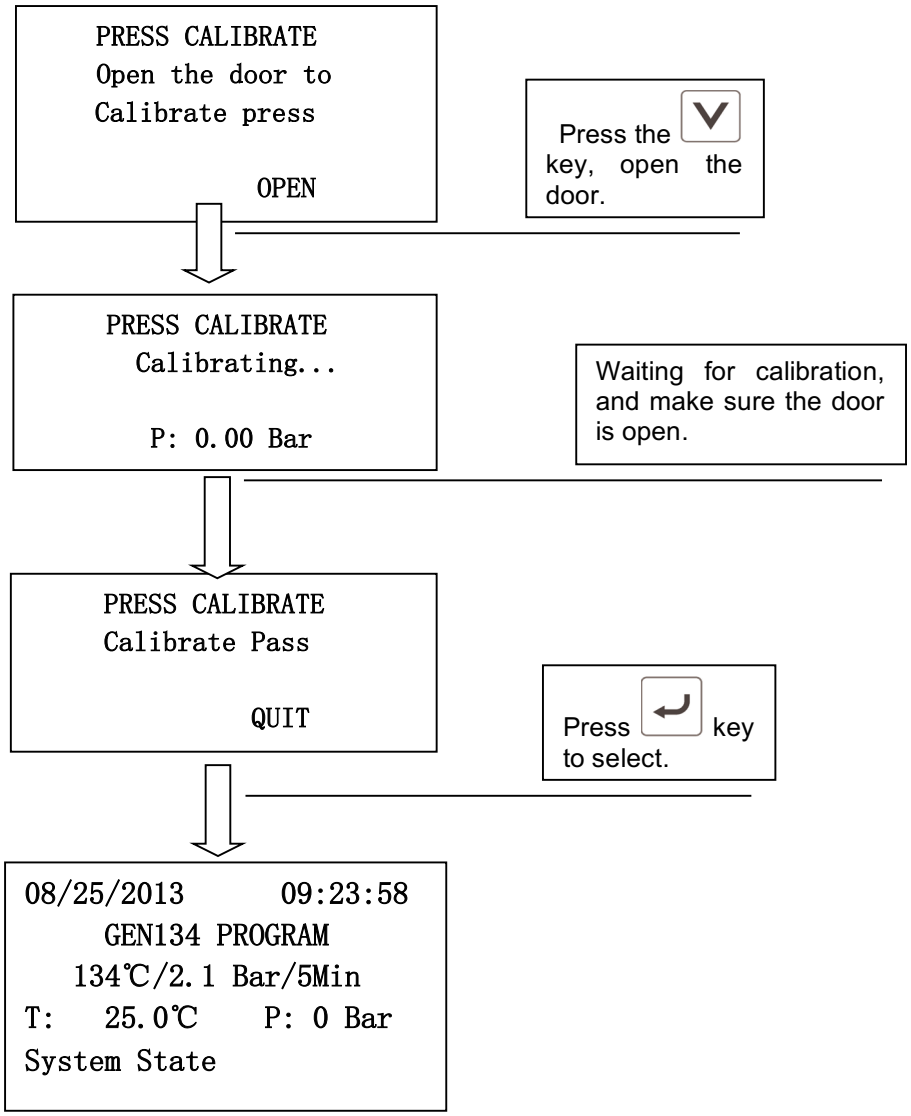
5 First-time Use

5.1 Printer (optional)

Sterilizer can be connected to an external printer to print the sterilization record and the fault messages generated in sterilization process. If printing is needed, the printer should be connected before starting the sterilizer. For details, please check the printer Installation and User Manual.

5.2 System Initialization

Users start system for the first time. Press the POWER switch to turn on the power of sterilizer. A HäTmed LOGO will show up, then atmosphere pressure calibration interface shows up. Operation procedures are as follows.



5.3 Vacuum Test

After first time installation, or transportation, or during regular examination, or in cases where the sterilizer has not being used for a long time, vacuum test must be carried out to examine the leakage of the sterilization system. It is preferred to do the vacuum test with a cooled system and dry chamber.

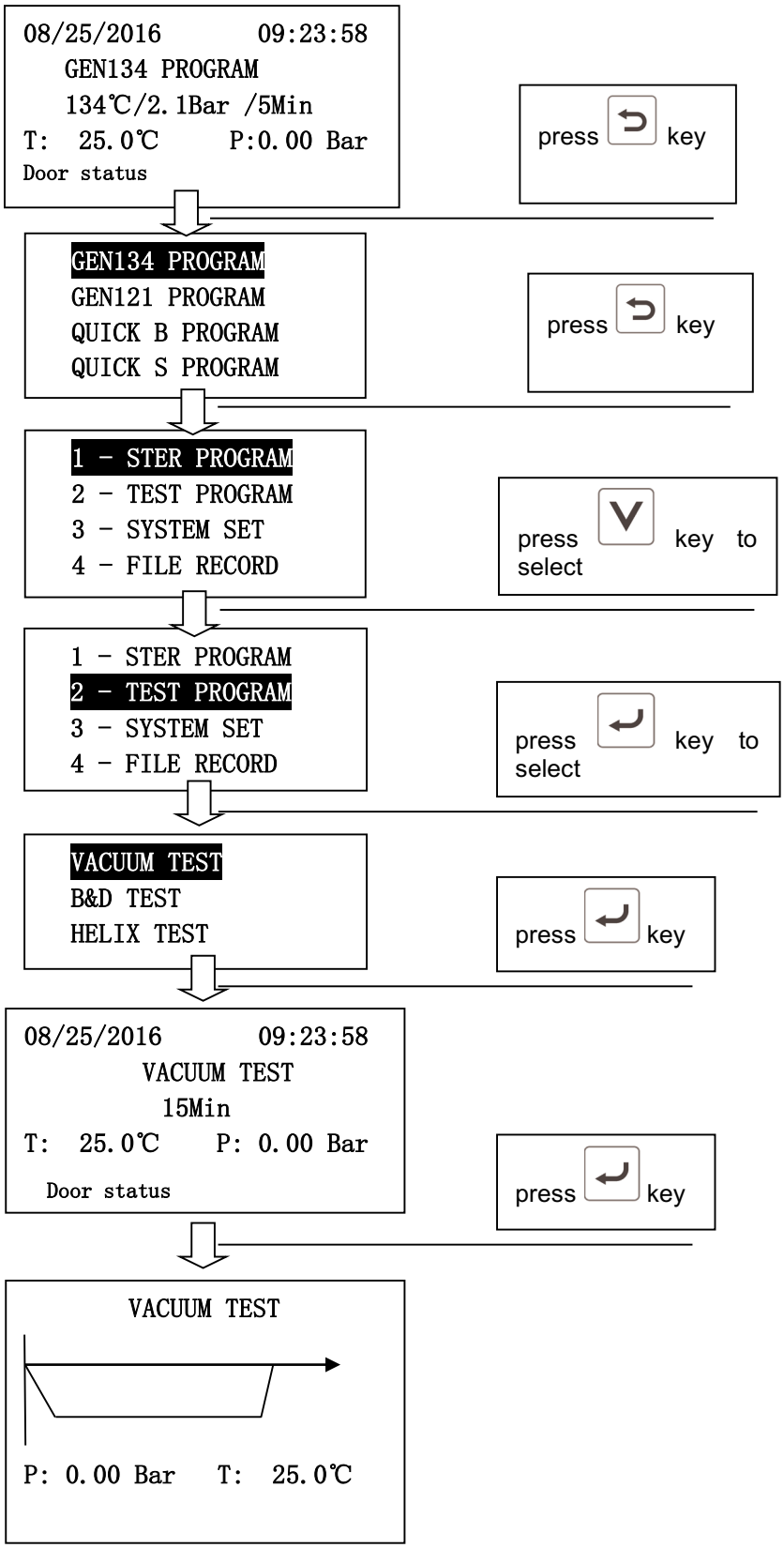
Vacuum test normally takes approximately 16 minutes, test results display on the screen after test finishes. When the screen shows test is successful, it indicates system is well sealed and there is no leakage. When the screen shows test fails, please do not use it and contact your distributor. If sterilizer connects to the printer, the information as follows can be printed after test is over.

- - - - - - - - - - - - - - - - Vacuum Test Pass! Leaking rate: <0.13Kpa/min - - - - - - - - - - - - - - - -	Vacuum test Passed! Leaking rate: <0.13Kpa/min
- - - - - - - - - - - - - - - - Vacuum Test Fail! Leaking rate: <0.13Kpa/min - - - - - - - - - - - - - - - - --	Vacuum test Failed! Leaking rate: <0.13Kpa/min

Note

- In order to complete the vacuum test smoothly, the door of the sterilizer must be locked. If the vacuum test fails, it indicates the system leakage does not meet the sterilization requirement. To ensure the quality of the sterilization, do not use the sterilizer before identifying the root cause.

Vacuum test operation process:



6 Sterilization Programs

6.1 Sterilization programs Overview

The sterilization programs include 6 specific programs. 6 specific programs are shown in Table 1:

Program	Tempera-ture(℃)	Sterilization time (min)	Drying time	Program time （min-max）	Sterilization load	Load capacity
Gen-121	121	16	15	60-90	A few textiles, heat-labile instruments, maximum 2kg textiles, or 6kg instruments.	2kg / 6kg
Gen-134	134	5.5	10	50-78	Mixed loads, complex hollow or porous devices. Packed or unpacked.	7kg
134+	134	5.5	10	63-98	Mixed loads, complex hollow or porous devices. Packed or unpacked.	7kg
Quick S	134	5	4	36-50	Simple solid metal pieces, geared instruments, instruments with simple straight cavity	6kg
Quick B	134	5	6	45-62	Maximum: 4kg packed devices, or 7kg unpacked simple solid metal piece	4kg / 7kg
Prion	134	20	10	55-82	Heat-resistant instruments that may be infected by Prion.	6kg
Textile	121	20	12	65-93	Textiles of 2.5kg.	2.5kg

Table 1 Parameters of the sterilization programs

6.2 Test Program

The 3 test programs are shown in Table 2 as follows:

Program	Tempera-ture (℃)	Pressure （Bar）	Sterilization time (min)	Maximum program time(min)	Sterilization load	Load capacity
Vacuum test	--	---	--	20	Empty load	0kg
Helix	134	2.1	5.5	60	Sterilization of the packed products, especially the instrument, or mixed load material (packed / unpacked)	7kg
B&D test	134	1.1	5.5	60	Sterilization of the packed products, especially the instrument, or mixed load material (packed / unpacked)	7kg

Table 2 Parameter of test Programs

7 Sterilization Procedure

7.1 Preparations

Check whether the connections to the sterilizer are proper, the door locked, distilled water supplied, printer connected, etc. Turn on the sterilizer, system will automatically check whether automatic pre-heating is selected, if set, system will start pre-heating program. Start sterilization program, system will automatically check whether the devices and peripherals mentioned above are effective. If they are effective, it will move on to the next stage. If any of them doesn't meet the requirements, sterilization will be stopped, and the screen will display fault messages and system will activate alarms.

7.1.1 Connect the Distilled Water

Distilled water should be correctly connected before starting the sterilization program. Connect the distilled water pipe from the distilled water supply to the distilled water inlet at the back of the sterilizer.

Note: In order to use the sterilizer smoothly, the quality of the distilled water must meet the requirements specified in EN13060.

7.1.2 Power Connection

Warning

- The power cord must be connected to a separate AC230V/16A power outlet correctly, and the power outlet must be placed in safe position.

7.2 Load the Sterilization Items

Loading the sterilization loads properly is the key factor for an effective sterilization and drying. When loading sterilization load, please be aware of the followings:

Warning

- Tray frame: The trays in the frame should be kept distance to each other.
- Tray: in order to make it easy for steam to distribute evenly in the loads, there should be holes on the tray. If the tray is solid, the sterilization load cannot be well sterilized and dried.
- Various packed items: The pulsating vacuum mode means that you can use a variety of packaging. But the load capacity should be less than the maximum load capacity of the selected program and the items should be kept distance to each other.
- Maximum load capacity: for instruments limit is 7 kg; for textile limit is 2.5 kg.

- Load capacity of each tray: for instruments limit 2 kg; for textile limit is 0.8 kg.
- Mixed sterilization load: put textile loads above and instruments.
- Make sure textiles do not touch the inner walls of the sterilizer chamber.
- The instructions for handling the instruments and textiles provided by their respective manufacturers must be followed.

7.3 Door Operation

Door opening: Press [V] in the main interface, the interface shows the door can be opened, then pull the door handle at the right side of the door to about 90° to open.

Door closing: Close sterilizer door, push the door handle downwards to the end.

7.4 Program Selection

The selected program must be suitable for the material characteristics of the corresponding sterilization load (especially heat resistant loads) and the type of the sterilization load.

Select appropriate program as below:

```
GEN134 PROGRAM
GEN121 PROGRAM
QUICK B PROGRAM
QUICK S PROGRAM
```

Press [∧] key or [V] key； press [↵] key to enter procedure introduction

7.5 Start the Program

Start the sterilization program as below, the screen displays the interface with runtime, temperature and pressure.

```
09/02/2016        09:23:58
      GEN134 PROGRAM
   134℃/2.1Bar/5Min
T:  25.0℃        P: 0 Bar
Door_status
```

Press [↵] key to start sterilize

7.6 Sterilization Program Procedure

After starting the sterilization program, the sterilizer will run automatically. The program will execute the following processes in turns:

Pre-heating

Activate the water suction pump to pump in distilled water. If the distilled water cannot flush in, maybe the distilled water pipes are stuck or leaking, sterilization program will be stopped, the screen will display fault messages and system will activate alarms.

Check whether pre-heating is on, turn it on if not. If it is hot start, the pre-heating could be very fast, if it is cold start, it requires a few minutes to preheat. The system will move on to the next stage after the pre-heating completes.

Vacuuming

Activate vacuum pump, the screen will display real-time pressure, temperature and the changing profiles of pressure and temperature. After vacuuming 3 times, activate steam generator to make the pressure rise to the sterilizing pressure, and then system moves on to the next stage.

Maintaining

It is also called sterilization stage, during which system maintains preset pressure and the preset temperature for a preset time duration, then the system moves on to drying stage.

Drying

Turn on the wastewater system to remove the residual moisture in the chamber, steam generators and pipelines, and turn on the vacuum system to start vacuum drying when the pressure is near the atmospheric pressure.

Completion

After drying stage completes, the system will extract fresh air into the sterilizer automatically, and prompt that sterilization is complete after reaching normal pressure. The system is working according to the configuration set by the user. If automatic printing is on and a printer is connected with the sterilizer, the system will print the sterilization record. At the same time, the record will be saved in the internal storage. If user wants to transfer the record to a computer, the record can be downloaded with a USB disk for long-term storage.

Warning

- Unpacked items should be used immediately, or stored in a sterile environment after sterilization, in order to prevent cross infection.

The sterilization records that can be printed output or stored in internal memory is formatted as below:

```
- - - - - - - - - - - - - - - - -
      Quantus Sterilizer
- - - - - - - - - - - - - - - - -
Program  : Quick-program-S
           134°C unwrapped
Data :        2017.07.12
Start Time :  09:11:29
Cycle :  1
SN :          65252120517050513
Preheat Temper :   95.0°C
Conductivity : 11.0us/cm
- - - - - - - - - - - - - - - - -
Program step    press  temper  time
                 bar     °C     min
 Start :       -0.04   69.6   0:00:00
1.Vacuum one
   Sub press : -0.87    72.0  0:02:26
  Steam press:  0.50    109.9 0:05:32
2.Vacuum two
   Sub press : -0.52    89.2  0:06:32
  Steam press:  2.14    134.3 0:11:51
- - - - - - - - - - - - - - - - -
Keep :          2.14   134.3  0:11:51
Keep End :      2.17   135.0  0:16:51
Realse :        0.10   105.6  0:18:27
Dry :           0.10   105.6  0:18:27
Dry End :      -0.91   90.3   0:22:27
Ventile :      -0.14   92.3   0:22:39
Finish :       -0.14   92.3   0:22:39
- - - - - - - - - - - - - - - - -
    PROGRAM PROPERLY EXECUTED!
Temperatur : 134.7+0.2/-0.4°C
Pressure : 2.14+0.02/-0.03bar
Sterilisierzeit :      0: 22:39
Uhrzeit End :          16:33:58
Hardware Version:      Qx3_V0.2
Software Version:      00.02.16.19

- - - - - - - - - - - - - - - - -
```

Sterilizer model

Program
Date
Start Time
Cycle #
Preheat temperature
Conductivity of distilled water

Twice vacuuming, record pressure, temperature and time

Heating up
Keep
Keep ends
Draining waste water
Drying beginning
Drying ends
Fill in fresh air
Sterilization finishes

Report, record pressure, temperature and time

7.7 Take out the Sterilization Load

Press key to open the sterilizer door after completion of sterilization. The screen prompts that door can be opened. Open the door and the sterilization load can be taken out.

Warning

- Be sure to take out the sterilization load with the tray handle. You may get scald by touching the chamber surface with bare hands.
- After taking out the packed sterilization load, check whether the package is damaged, for example, seal is broken. If so, the sterilization load should be repacked and re-sterilized.
- If the sterilization is not interrupted and the sterilization load is placed correctly, the sterilization load can be dried well. After the sterilization, there can be residual condensates on the sterilization loads or trays, usually these condensates will evaporate soon after sterilization.

7.8 Print the Records

If the printer is connected correctly and automatic printing is preset (how to setting automatic printing is described in 8.2.3), the sterilization record will be printed automatically.

7.9 Terminate the Program Manually

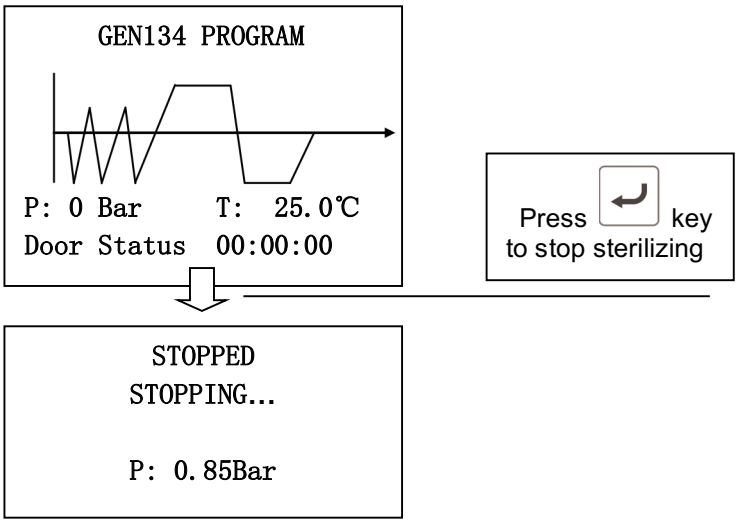
7.9.1 Terminate the Sterilization

After starting the sterilization program, you can terminate the sterilization at any time.

To terminate the sterilization, the specific steps as below. After the sterilization stops, system will take appropriate actions accordingly and ensure a safe and proper state is reached before allowing user to open the door.

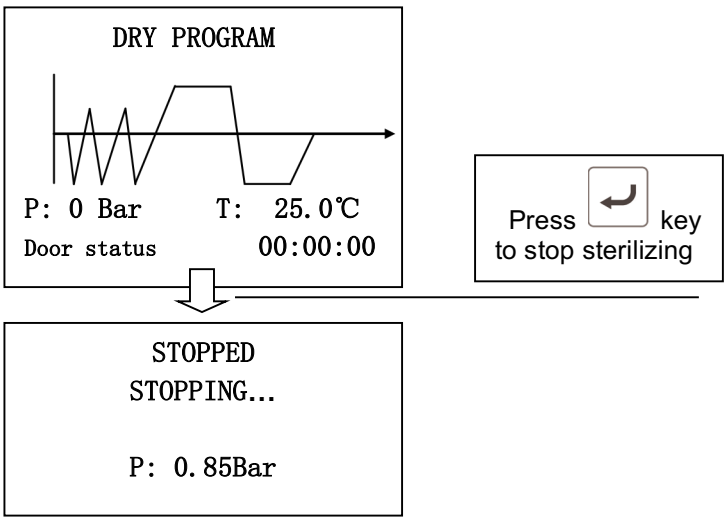
Warning

- Please note that hot steam may come out when opening the door after you terminate the program, depending on the time of the termination.
- If the program is terminated before the beginning of drying phase, then the items to be sterilized are still unsterile, and the sterilization is NOT successful.
- If the sterilization is manually terminated, in order to ensure the quality of the next sterilization and reduce the sterilization time, it is suggested to sterilize with no load once before the next sterilization.



7.9.2 Terminate the Drying

The program may be terminated in the drying phase of the sterilization program to save time; the sterilization has already been completed, so the sterilization load has already been sterilized. The sterilization load will not be dry sufficiently, especially for the packed items. If possible, let the programs with packed items to be sterilized continue until the end of the drying phase.



8 Special Functions

8.1 Sterilization Documentation

8.1.1 Introduction of Sterilization documentation

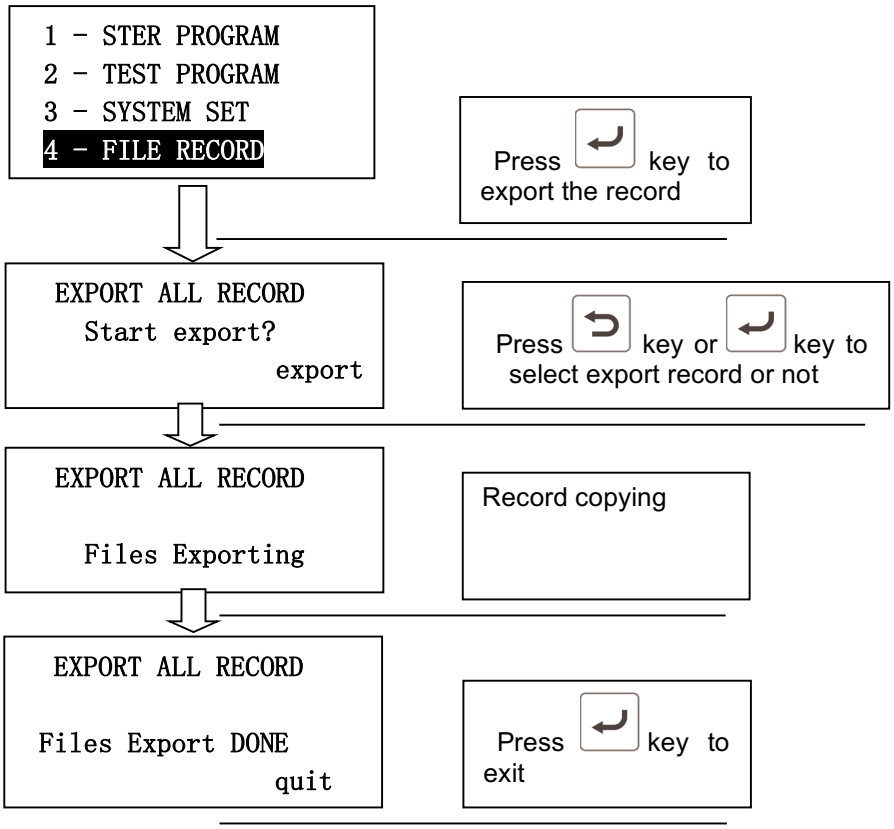
The sterilization logs documented real-time information of the sterilization process. It is stored in the sterilizer.

The log name is named after cycle number. For example '001.txt' is the record of the first cycle. This format ensures that there will not be duplicated names, and it is convenient for search and store.

Sterilization records can be exported through the USB interface or printed out via a printer.

8.1.2 Sterilization Log Export

The sterilization logs can be exported through the USB interface to a computer or other recording device. Operation steps for USB disk are as follows: Insert the USB disk into the USB interface, then follow the procedure as shown below to export the sterilization record. When the screen displays the information that export is finished after exporting, the USB disk can be removed.



8.1.3 Printing the Sterilization Record

8.1.3.1 Printer Connection

Warning

- Printer data line should be connected to the printer connector at the back of the sterilizer （
- Figure 2-4）, between the DB9 interface and the DB9 socket at the back of the printer, then tighten the screws.
- Use power supply and power cord provided with the printer package.
- Turn on the power at the back of the printer, the power indicator (green light) is on, the status indicator (red light) is off. The red light will flag when lack of printing paper.
- More operation please refer to the operation manual of the printer.

8.1.3.2 Printing Sterilization Log

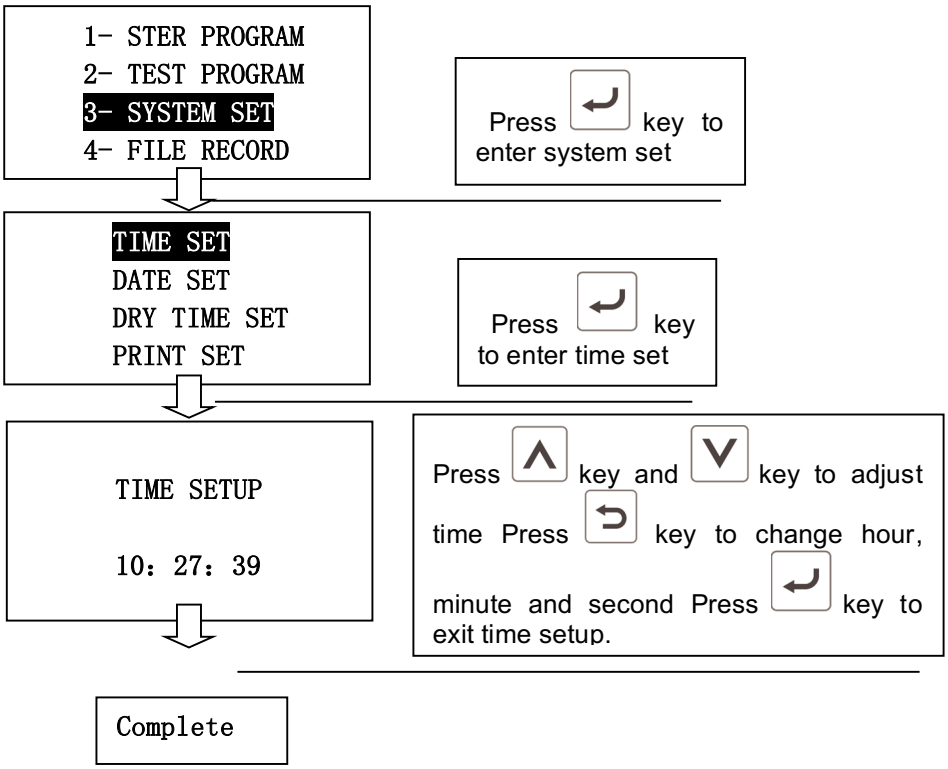
If the automatic printing function is on, sterilization record is printed automatically every time after sterilization completes.

8.2 System Configuration

Time, date, drying time, printer and system reset of sterilizer can be configured.

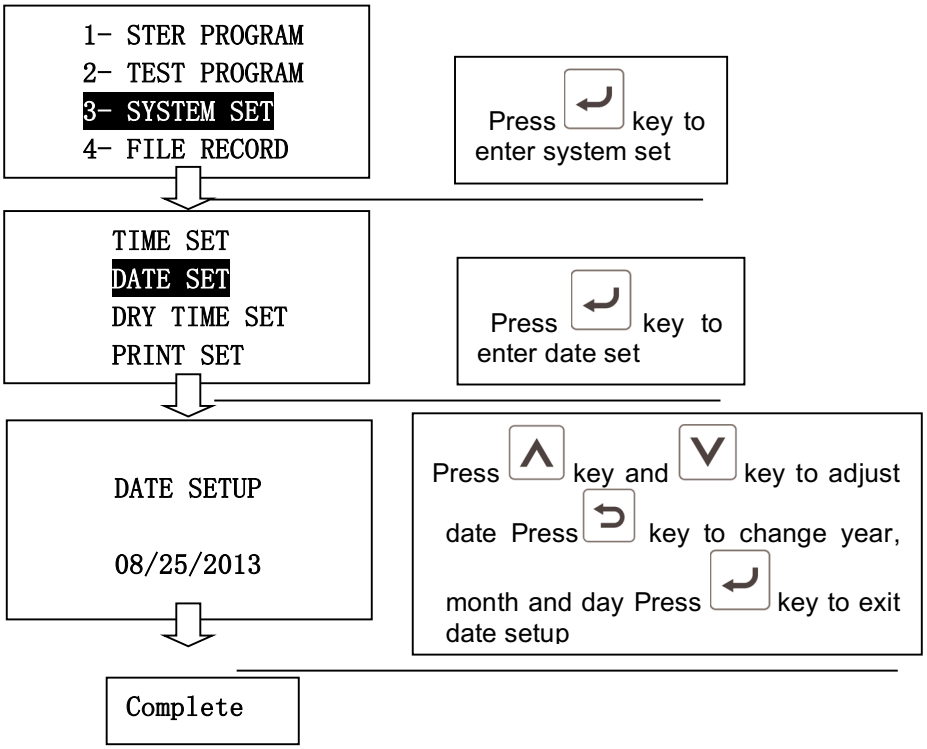
8.2.1 Time Setting

Set time as below:



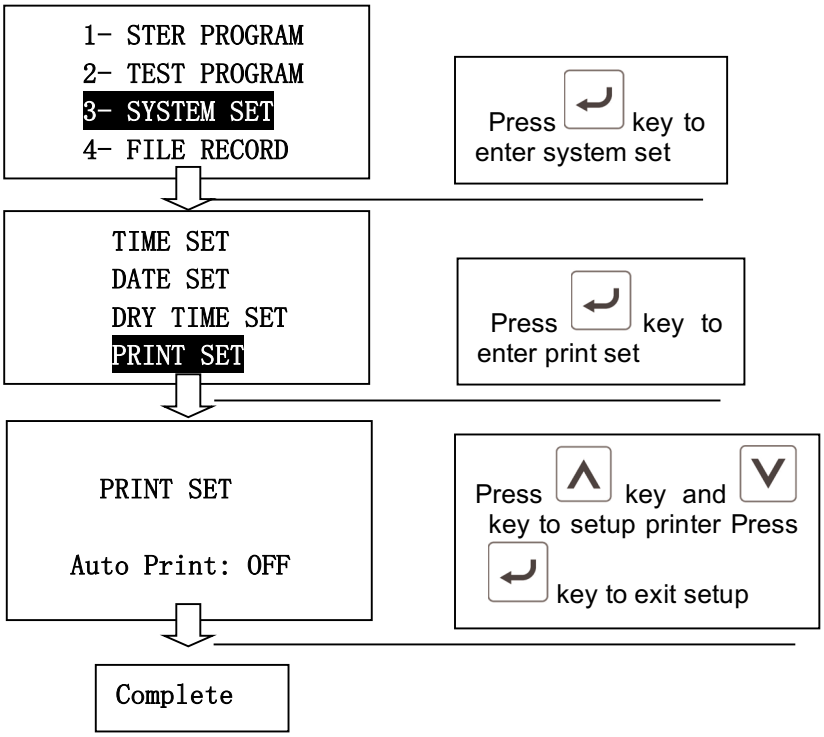
8.2.2 Date Setting

Set date as below:



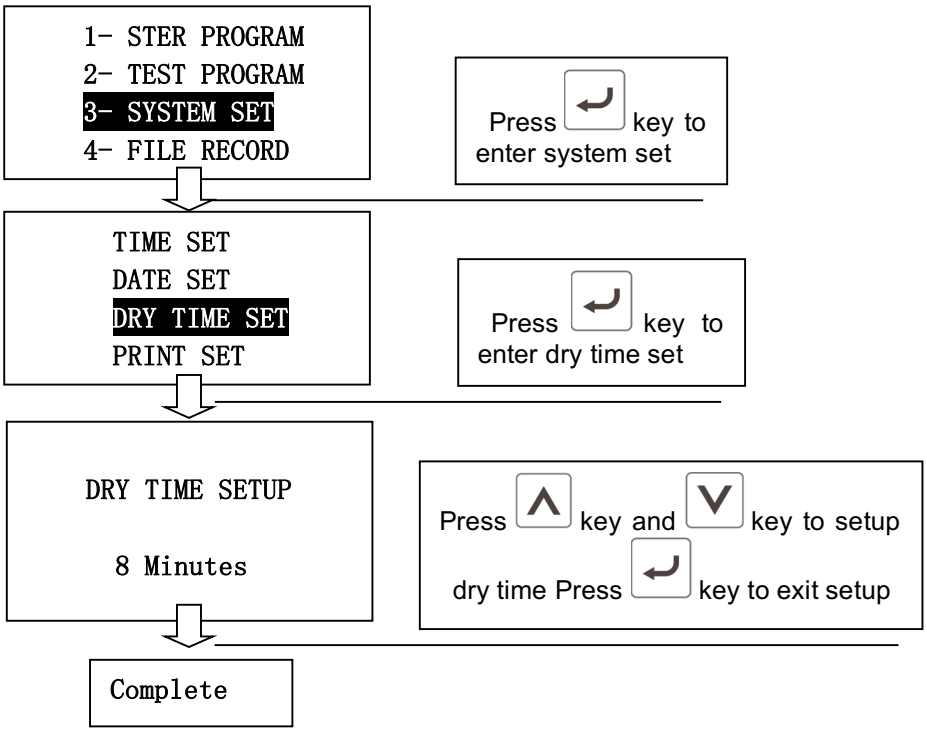
8.2.3 Printer setting

Set printing option as below:



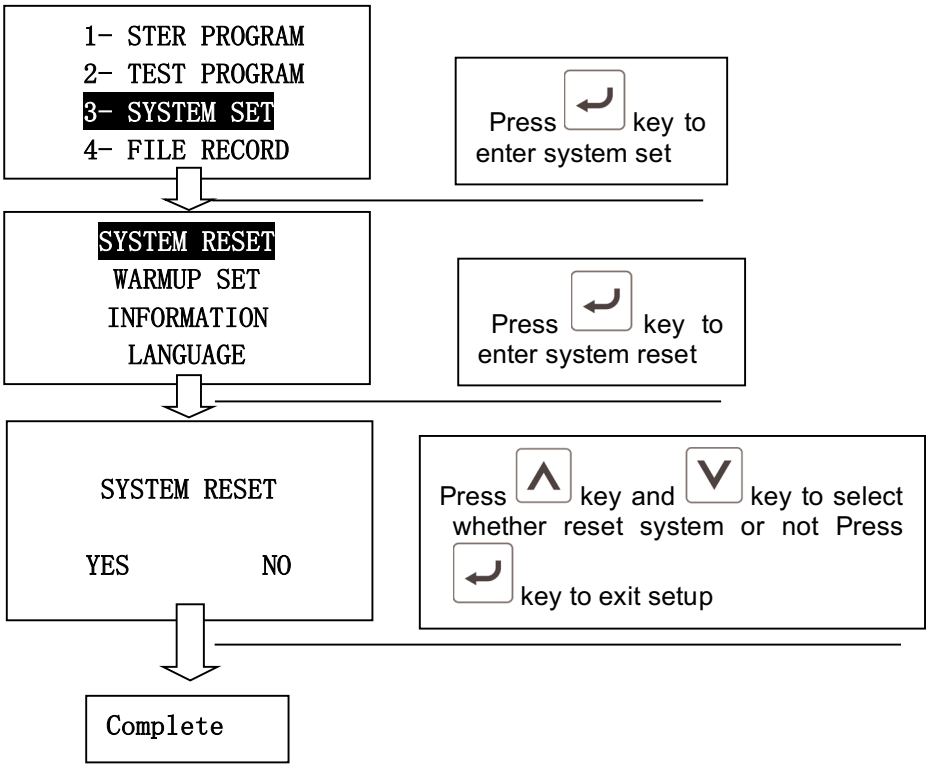
8.2.4 Drying time setting

Set drying time as below:



8.2.5 System reset

System reset as below:



8.3 Additional drying Program

This extra drying program guarantees very good drying result for items to be sterilized. Additional drying program is not necessary in most cases. A typical use of this program is to dry a large amount of textile load after the sterilization program.

9 Warnings and Error Messages

9.1 Warning Messages

The sterilizer prompts warning messages whenever necessary to give notice and alert of an erroneous situation. An example of a warning message: “No distilled water”. When warning messages prompt, the sterilizer also gives alarm sound. All warning messages are listed below:

Code	Warning Message	Causes	Checking or Maintenance Approaches
	Please close the door	The door is not closed correctly.	Close the door and push the door handle towards the end
	Please fill the tank	Water level in the tank is too low; Distilled water is not supplied smoothly.	1、Check water level in the tank 2、Check if the filter at the bottom of the water tank is stuck, and clean it accordingly 3、Check flow meter
	Please drain the tank	Water level in the waste water tank is too high	1、Drain water from the waste water tank.
	Cool the chamber	Temperature in chamber is higher than 45°during vacuum test	1、Open the door; start vacuum test after the chamber is cool.
	High Pressure	Pressure in chamber >1.04Bar	1、Wait until pressure in chamber becomes atmosphere pressure 2、Contact your distributor if pressure doesn’t become normal
	Low Pressure	Pressure in chamber <-0.95Bar	1、Wait until pressure in chamber becomes atmosphere pressure 2、Contact your distributor if pressure doesn’t become normal
	NO USB Disk	USB connection error	1、Check if USB cable is connected 2、Contact your distributor if the warning prompts repeatedly
	NO Record	No log is found. It means the sterilizer has not run any program or just gone through system reset.	1、Run one sterilization cycle
02	Forced Stop	Stop button was pushed when during sterilization cycle	1、Re-run the sterilization cycle after checking the device
03	Door Sensor 1	Door sensor status is open during sterilization cycle	1、Check the door sensors and make sure door is closed tightly and door handle pushed to the end. 2、If same warning prompts repeatedly, contact your distributor.
04	Door Sensor 2	Door sensor status is open during sterilization cycle	1、Check the door sensors and make sure door is closed tightly and door handle pushed to the end. 2、If same warning prompts repeatedly,

			contact your distributor.
05	Door Sensor 1&2	Door sensor status is open during sterilization cycle	1、Check the door sensors and make sure door is closed tightly and door handle pushed to the end. 2、If same warning prompts repeatedly, contact your distributor.
06	Preheat Timeout	Heater fault	1、Check the heater wrapped around the chamber 2、If same warning prompts repeatedly, contact your distributor.
08	No Dist Water	Distilled water pipe is stuck/flow meter signal timeout	1、Check if distilled water pipe is twisted or knotted； 2、Check flow meter 3、If same warning prompts repeatedly, contact your distributor.
09	Water Quality	Unqualified distilled water	1、Use better quality distilled water 2、Clean water tank regularly
11	Vacuum Timeout	Vacuum test failure	1. Check and clean seals; 2. Run vacuum test with cool and dried chamber 3. If the message appears repeatedly, please contact your distributor. **Warning** Please do not adjust the gate spacing without guide from qualified personnel. When vacuum test fail, do not perform any operation using the sterilizer, please contact your distributor.
13	Steam Timeout	Steam Generator Fault	1、Check the steam generator 2、If same warning prompts repeatedly, contact your distributor.
16	Temper SW Open	Temperature switch open	1、Check temperature switch and other cable connection 2、If same warning prompts repeatedly, contact your distributor.
17	Release Timeout	Exhaust valve fault	1、Check the exhaust valve 2、If same warning prompts repeatedly, contact your distributor.
20	System Error	System configuration error	1、Check system configuration 2、If same warning prompts repeatedly, contact your distributor.

22	Drain Timeout	Draining valve fault	1、Check draining valve 2、If same warning prompts repeatedly, contact your distributor.
23	Levelling Timeout	Air filter valve fault	1、Check air filter valve 2、If same warning prompts repeatedly, contact your distributor.
24	Press Max Limit	Pressure in chamber exceeds upper limit during operation	1、Stop the device, check the device thoroughly，if no other error，try again 2、If same warning prompts repeatedly, contact your distributor.
26	Press Min Limit	Pressure in chamber exceeds lower limit during operation	1、Stop the device, check the device thoroughly，if no other error，try again 2、If same warning prompts repeatedly, contact your distributor.
25	Temper Max Limit	Temperature in chamber exceeds upper limit during operation	1、Stop the device, check the device thoroughly，if no other error，try again 2、If same warning prompts repeatedly, contact your distributor.
27	Temper Min Limit	Temperature in chamber exceeds lower limit during operation	1、Stop the device, check the device thoroughly，if no other error，try again 2、If same warning prompts repeatedly, contact your distributor.
30	Balance Timeout	Leaking	1、Check pipelines, door sealing, chamber opening for leakage

10 Troubleshooting

10.1 Screen No Display

Situation: No display or abnormal display after power on the sterilizer.

Solution:

- ✧ Check whether the power outlet has power;
- ✧ Check whether the power plug of the sterilizer is inserted;

10.2 Large Consumption of Distilled Water

Situation: The consumption of distilled water is larger than normal.

Solution:

- ✧ Check the foot pads, the front ones should be higher than the rear ones;
- ✧ Check whether the filter at the bottom of the chamber is stuck;

10.3 Load not dried sufficiently

Situation: Loads are not dried sufficiently.

Solution:

- ✧ Check the foot pads, the front ones should be higher than the rear ones;
- ✧ Check whether the filter net at the bottom of the chamber is stuck;
- ✧ Check whether the maximum load capacity is exceeded (especially textile), whether the sterilization load is placed correctly and allows easy dissipation of steam.
- ✧ Activate the preheating function;
- ✧ Open the left side panel, check if the filter is stuck;

10.4 No Distilled Water

Situation: No Distilled Water.

Solution:

- ✧ Check if the distilled water tank has sufficient water in it;
- ✧ Check if the distilled water pipe is twisted or knotted;

10.5 Leaking air or water at the door opening

Situation: Observable air leaking or water drops at the door opening.

Solution:

- ✧ Check if the sealing ring is placed corrected (Wider side with “H” mark towards the chamber);
- ✧ Check if the sealing ring is broken;

If the solutions above do not work, please contact HäTmed distributor.

11 Maintenance

11.1 Preparation of the Sterilization Load before Sterilization

Warning

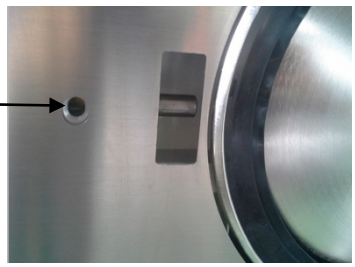
- The sterilization loads such as handpiece should be properly cleaned before sterilization.
- It is recommended to clean the sterilization loads such as scissor by washing with cleaning solutions or by ultrasonic cleaning according to manufacturers' specifications.
- In order to prevent clogging of the filters, valves and pipes from the dust and other contaminants separated from the sterilization loads, the cleaning of the sterilization loads before sterilization is very important. Therefore, the sterilization loads without cleaning are not allowed to be sterilized in the chamber.
- Follow the requirements provided by the instrument manufacturer.

11.2 Maintenance

11.2.1 Cleaning

After having been used for a certain period, some parts of the sterilizer should be cleaned or checked.

Maintenance	Maintenance period	Tool	Operation
Replace the seal ring	1000 sterilization cycles		Open the door, check whether the seals is deformed or wrinkled, and replace it if so.
Sealing ring Maintenance	Two weeks		Take off the sealing ring, use wet clean cloth to clean the sealing ring inside out. Then use dry clean cloth to dry the sealing ring. Put the sealing ring back to the sterilizer. Note: the narrower side of the sealing ring should be close to the lid and the wider side of the sealing ring towards to chamber

Lock Maintenance	100 sterilization cycles	Lubricating oil	Spray some oil into the “A” spot at the left side of the lid, clean the outside with a rag. 
Clean the distilled water tank	300 sterilization cycles		Wash it with clean water
Clean the filter at the bottom of the chamber	200 sterilization cycles	Flat screw driver	Remove the filter with a wrench, and wash it with clean water.
Clean the trays	100 sterilization cycles		Wash it with clean water, use toothpaste to clean the stains.
Door Shaft	Two weeks	Lubricating Oil	Spray few drops of lubricating oil
Sterilization Chamber	One week		Clean and try the chamber with medical cotton cloth

Warning

- If maintenance or repair is needed, contact HäTmed distributor.
- Maintenance and repair of the sterilizer must be carried out by qualified personnel.

11.2.2 Regular Maintenance

In order to ensure sterilization quality and your safety, maintenance is recommended when the system prompts that the sterilization times have exceeded the preset limit.

11.2.3 Safety Mechanism

In order to ensure that the system is safe and reliable, a series of protective mechanism are used to protect the critical components.

The preheating system over-temperature protection switch: There is an over-temperature protection switch in the preheating system. When the temperature exceeds the threshold of the over-temperature protection switch, the power supply circuit of the preheating system will be automatically shut down to protect the preheating system.

The steam generator over-temperature protection switch: There is an over-temperature protection switch under the steam generator. When the temperature of the steam generator exceeds the threshold of the over-temperature protection switch, the power supply circuit of the steam generator will be automatically shut down to protect the steam generator.

The safety valve: When the pressure in the chamber reaches 1.3 times of working pressure, the safety valve opens automatically to keep the pressure in safe range.